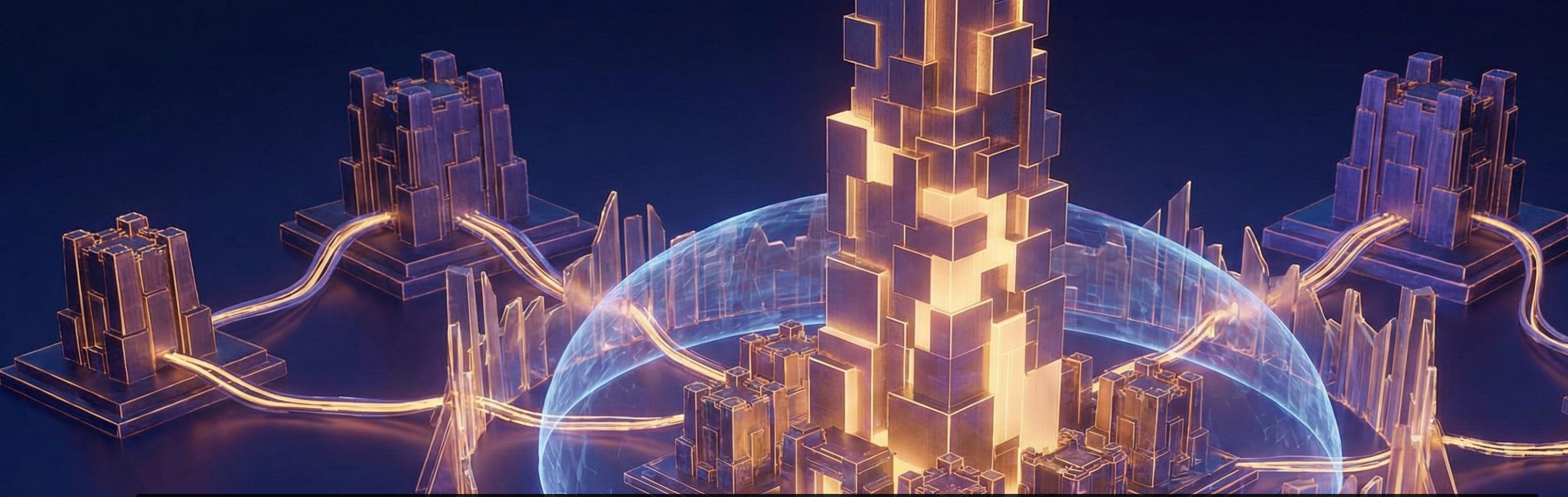


# The Competitive Landscape – Market Structure and Monopoly

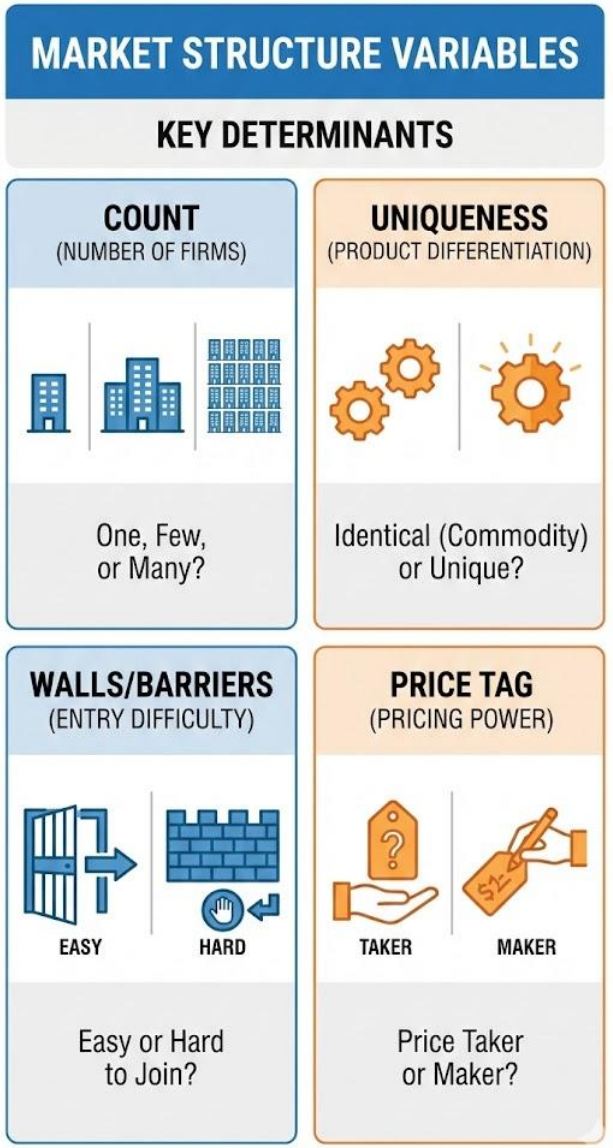
## Business Competition, Market Structure, Monopoly, Oligopoly, Barriers to Entry

**ITGE275 Economics**

Associate Prof. Dr. Suppawong Tuarob
Faculty of ICT, Mahidol University

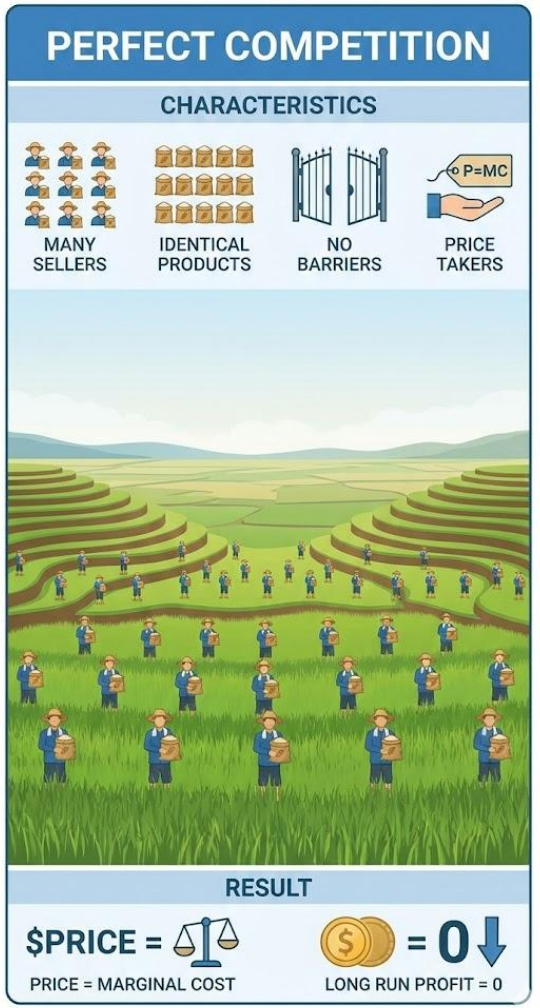
# Defining Market Structure

- **Definition:** Market structure refers to the organizational characteristics of a market that determine the behavior of firms and the interaction between buyers and sellers.
- **Key Variables:**
  - **Number of Firms:** One, Few, or Many?
  - **Product Differentiation:** Identical (Commodity: crude oil, gold, or wheat, etc.) or Unique?
  - **Barriers to Entry:** Is it easy or hard for new startups to join?
  - **Pricing Power:** Is the firm a Price Taker or a Price Maker?



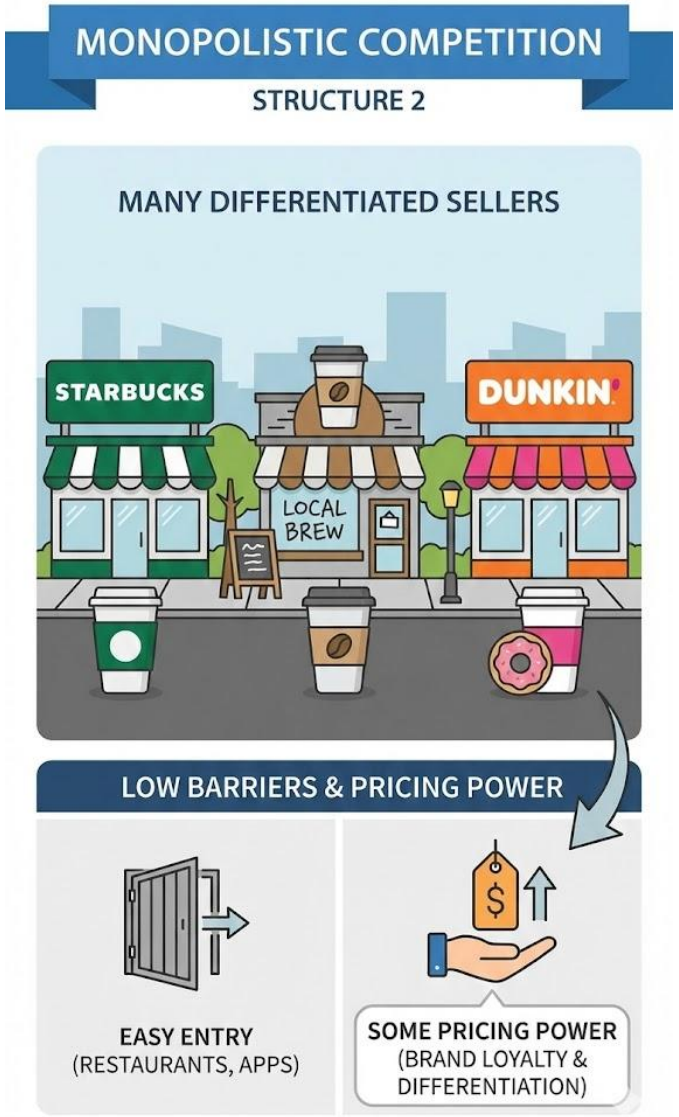
# Structure 1: Perfect Competition

- **Characteristics:**
  - **Many Sellers:** Thousands of small firms.
  - **Identical Products:** Consumer cannot distinguish between Supplier A and Supplier B (e.g., Grade A Jasmine Rice).
  - **No Barriers:** Anyone can start selling tomorrow.
- **Result: Price Takers.** No single firm can influence the market price. ***Price = Marginal Cost***.
- **Profit:** In the long run, economic profit is driven to zero.



# Structure 2: Monopolistic Competition

- **Characteristics:**
  - **Many Sellers:** High competition.
  - **Differentiated Products:** Products are similar but not identical (e.g., Restaurants, Clothing, Mobile Apps).
  - **Low Barriers:** Easy to open a restaurant or launch an app.
- **Result:** Firms have **some** pricing power based on Brand Loyalty and Differentiation.



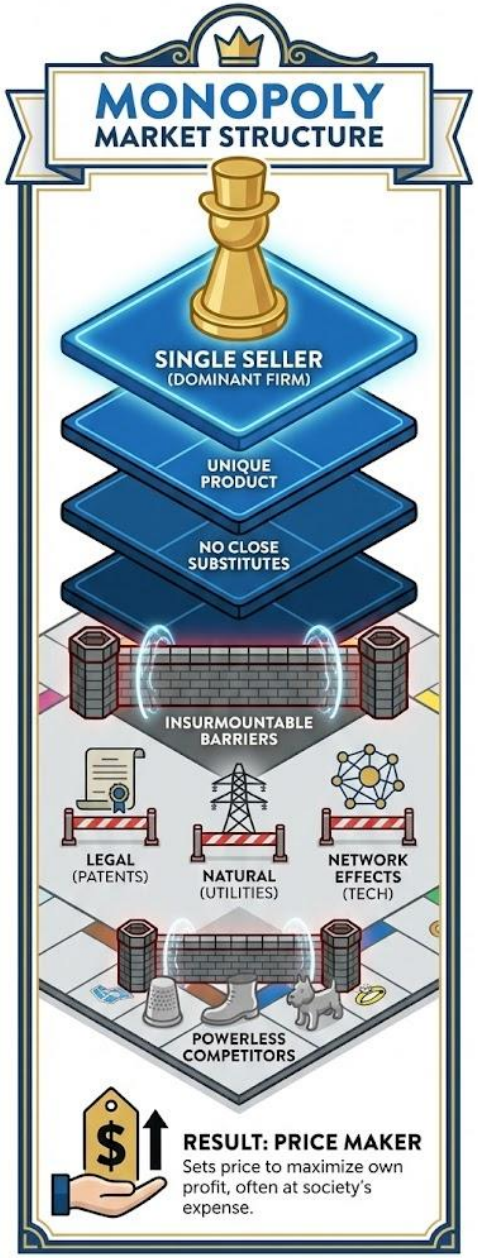
# Structure 3: Oligopoly

- **Characteristics:**
  - **Few Dominant Sellers:** 2-5 firms control 80%+ of the market.
  - **High Barriers:** Massive capital requirements (e.g., Building cell towers, buying airplanes).
  - **Interdependence:** Firm A's decision significantly impacts Firm B (Game Theory applies).
- **Examples:** Telecoms (AIS/True), Operating Systems (iOS/Android), Aircraft (Boeing/Airbus).



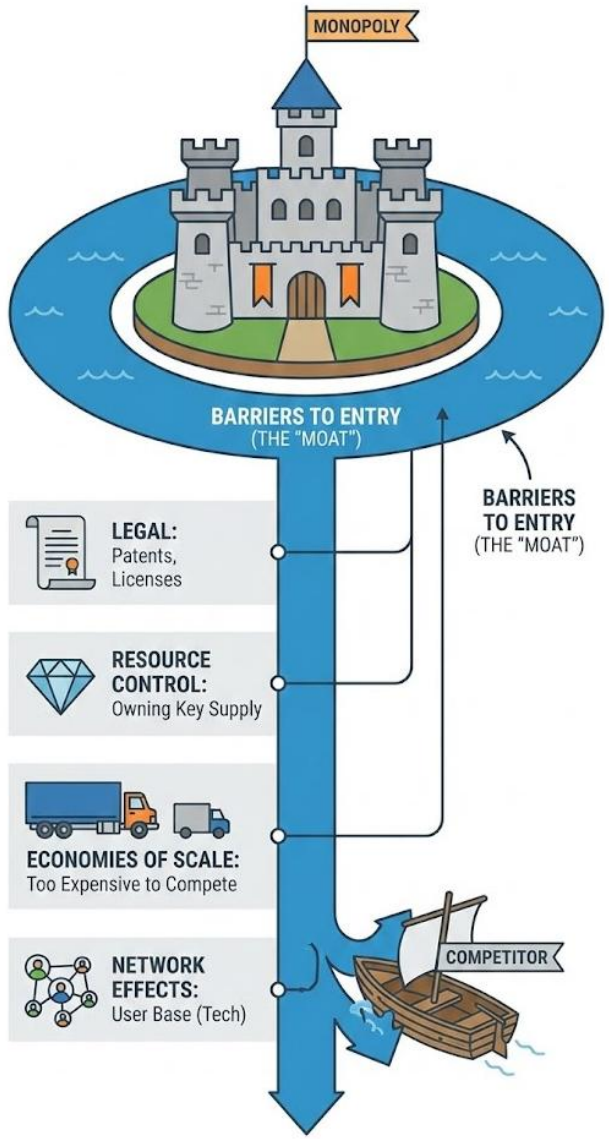
# Structure 4: Monopoly

- **Characteristics:**
  - **Single Seller:** One firm controls the entire market.
  - **Unique Product:** No close substitutes.
  - **Insurmountable Barriers:** Legal (Patents), Natural (Utilities), or Network Effects (Tech).
- **Result: Price Maker.** The firm sets the price to maximize its own profit, often at the expense of society.



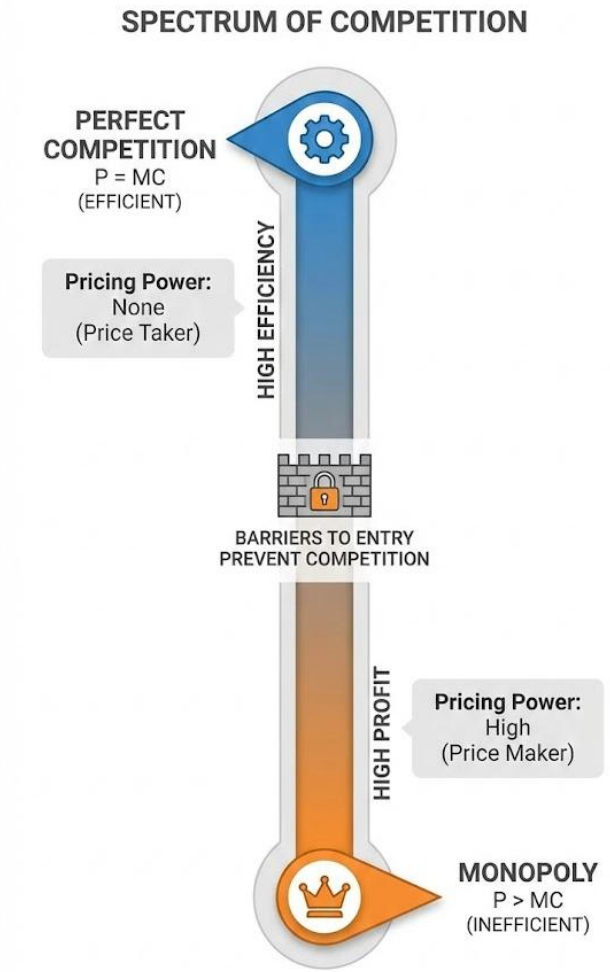
# Barriers to Entry (The "Moat")

- Why don't competitors just enter and lower the price? **Barriers.**
- **Types of Barriers:**
  - **Legal:** Patents, Copyrights, Government Licenses (e.g., Pharma, Utilities).
  - **Resource Control:** Owning the only diamond mine.
  - **Economies of Scale:** It is too expensive for a small startup to compete with Amazon's logistics.
  - **Network Effects (Tech):** A social network is useless without users. The incumbent has all the users.



# Part 1 Summary

- The "Spectrum of Competition" dictates pricing power.
- **Perfect Comp:** P = MC (Efficient).
- **Monopoly:** P > MC (Inefficient).
- **Barriers to Entry** prevent the market from becoming competitive.
- **Next:** How does a Monopoly actually set prices?

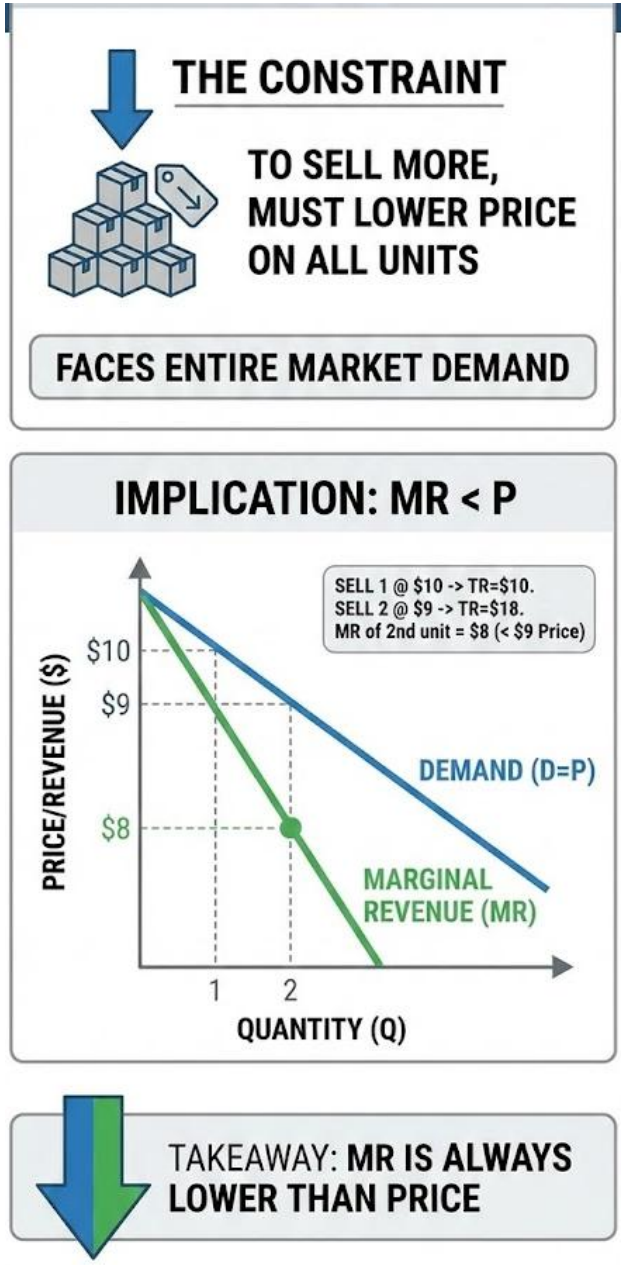


## Part 2: Monopoly Mechanics & Welfare

# The Monopolist's Dilemma

- Unlike a competitive firm, a Monopolist faces the **entire Market Demand Curve**.
- **The Constraint:** To sell *more* units, the Monopolist must *lower* the price on ALL units.
- **Implication*:** Marginal Revenue (MR) is always **lower** than Price (*P*).
  - If you sell 1 unit at $10, Revenue = $10.
  - To sell 2 units, you must drop price to $9. Revenue = $18.
  - Marginal Revenue of 2nd unit = $8 (which is less than the $9 price).

*The firm faces a downward-sloping market demand curve, requiring it to lower the price on all units sold—not just the additional unit—to increase sales volume. This "price effect" reduces revenue from previous units, making the marginal revenue gain less than the price charged.

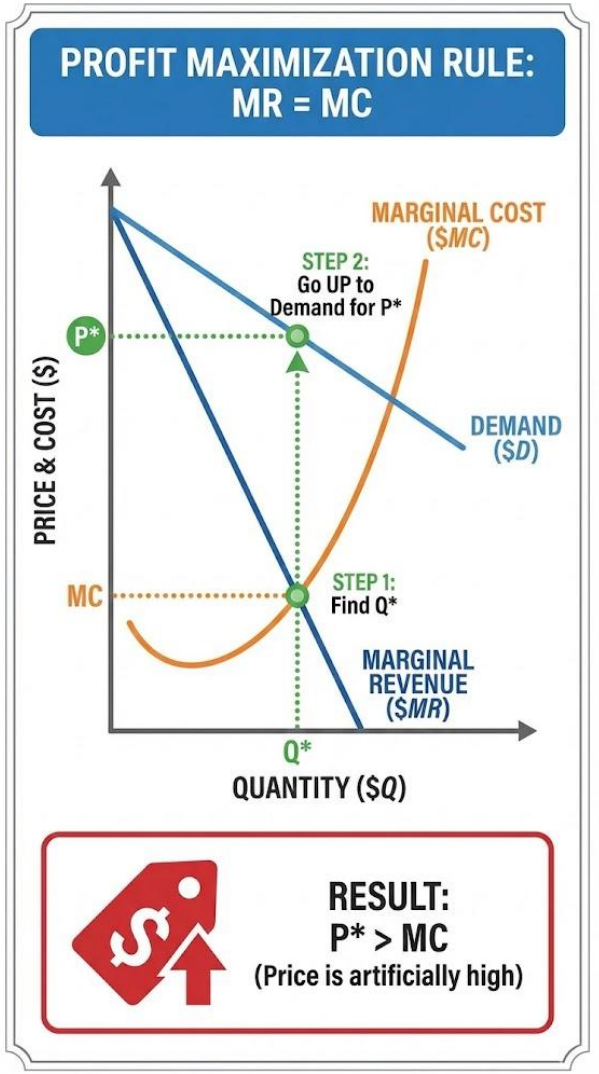


# Profit Maximization for Monopolies

- **Rule:** Produce where

  ***Marginal Revenue (MR) = Marginal Cost (MC)***

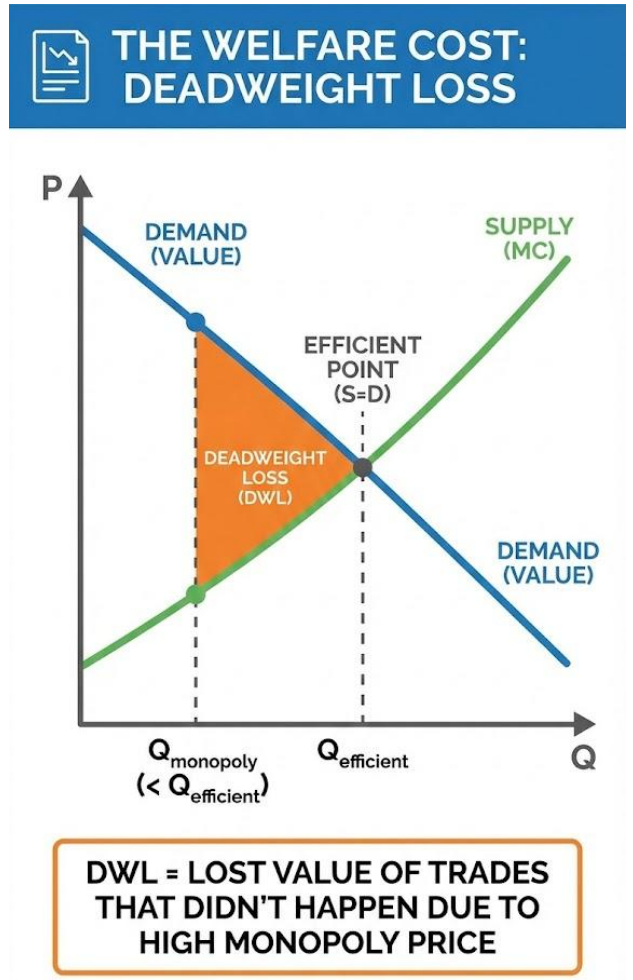
- **The Process:**
  - Find Q* where ***MR*** intersects ***MC***.
  - Go **UP** to the Demand Curve to find the Price (***P****) consumers are willing to pay for that quantity.
- **Result: *P* > MC***. The price is artificially high.



# The Welfare Cost (Deadweight Loss)

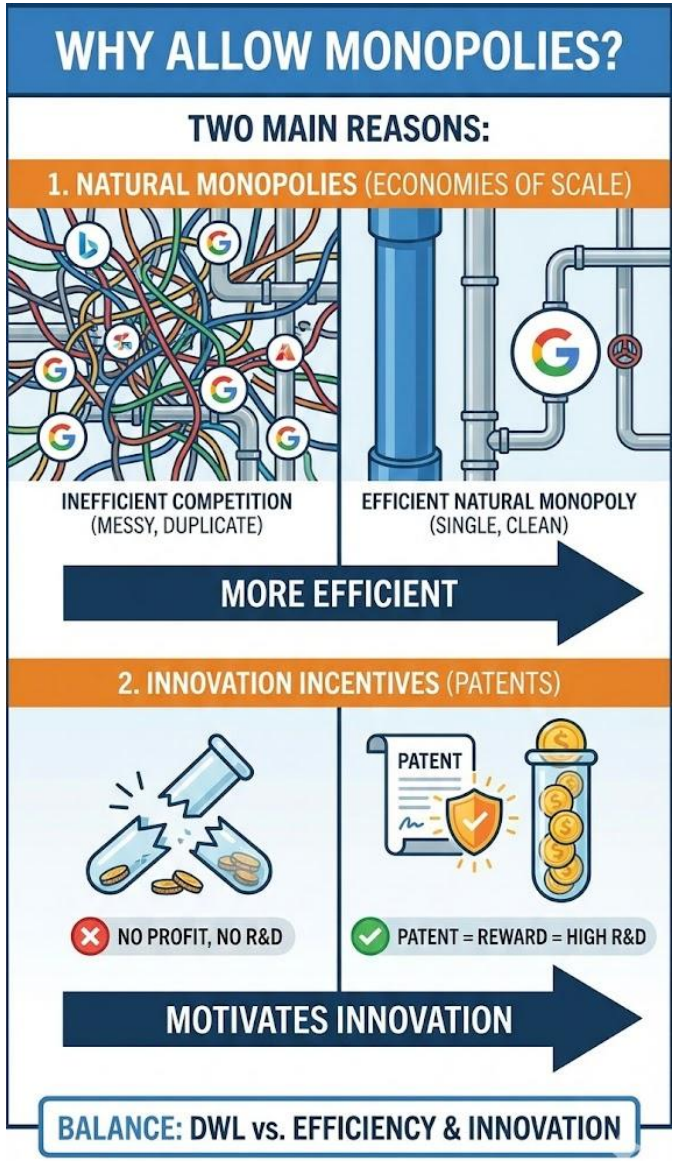
- **Efficiency:** Occurs where Supply (MC) equals Demand. This maximizes the total value to society.
- **Inefficiency:** Monopolies restrict output ($Q_{monopoly} < Q_{efficient}$) to keep prices high.
- **Deadweight Loss (DWL):** The lost value of trades that *should* have happened but didn't because the price was too high. It is a loss to society that no one captures (not even the monopolist).





# Why do we allow Monopolies?

- If they cause Deadweight Loss, why are they legal?
- **1. Natural Monopolies:** It makes no sense to have 5 different water pipe companies digging up the street. One firm is more efficient (Economies of Scale).
- **2. Innovation Incentives:** Patents (temporary monopolies) reward inventors. Without the promise of monopoly profit, Pharma might not invest billions in R&D.



# Antitrust & Regulation

- Governments police markets to prevent abuse of power.
- **US:** Department of Justice (DOJ), FTC.
- **Thailand:** Trade Competition Commission (TCCT).
- **Tools:**
  - Blocking Mergers (e.g., Preventing two giants from combining).
  - Breaking up companies (e.g., Standard Oil, AT&T).
  - Fines for anti-competitive behavior.



# Tech Giants & Antitrust (Case Study)



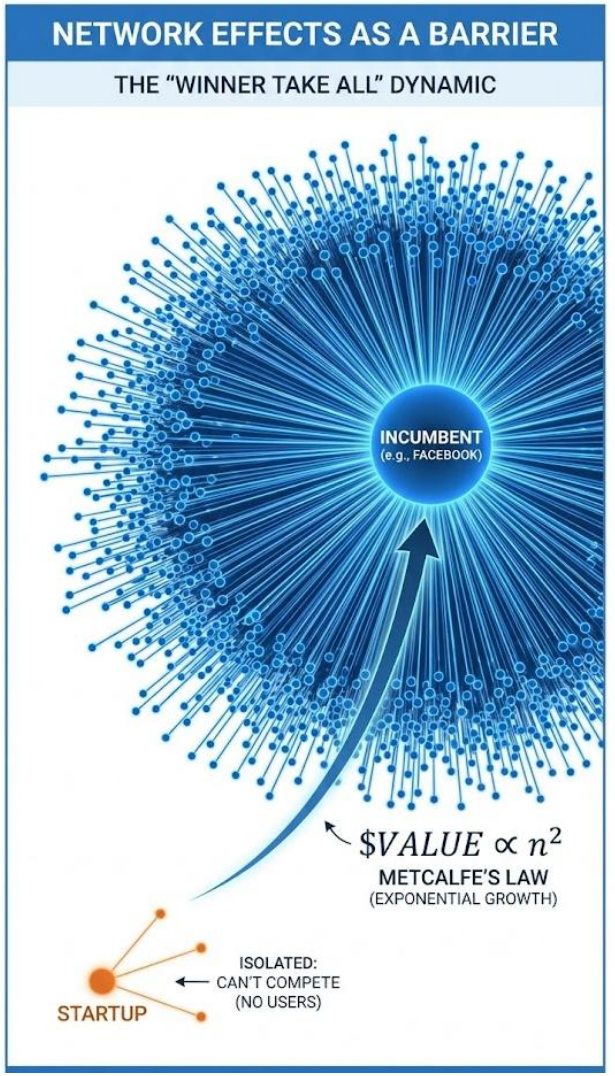

- **The Microsoft Case (1990s):** Microsoft bundled Internet Explorer with Windows.
  - *Accusation:* Using monopoly in OS (Windows) to destroy competition in Browsers (Netscape).
- **Modern Parallels:**
  - **Google:** Search dominance & Ad tech.
  - **Apple:** App Store fees & "Walled Garden.“*
  - **Amazon:** Using data from 3rd party sellers to compete against them.

*Apple’s "walled garden" refers to its tightly controlled iOS ecosystem, where the App Store is the exclusive, curated channel for software, enforcing high security but restricted competition. Apple charges developers a 15–30% commission on digital sales, a fee structure currently under legal pressure from antitrust rulings that allow alternative payment methods and app stores.

https://www.paddle.com/news/industry/apple-app-store-developers-bypass-fees-spotify-delta-patreon

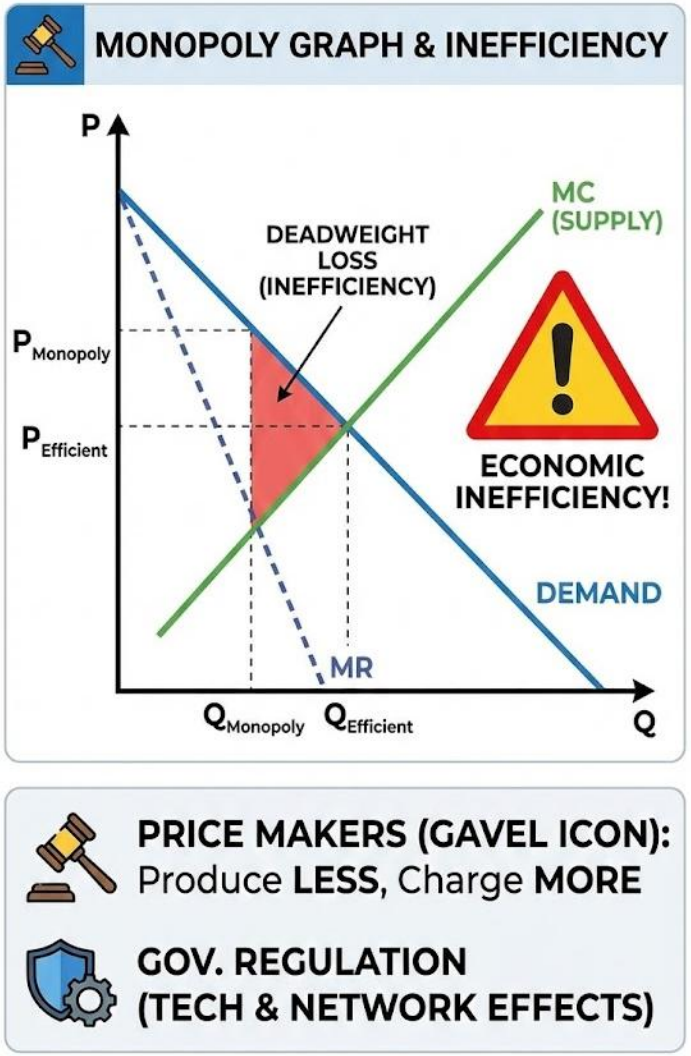
# Network Effects as a Barrier

- In Tech, the strongest barrier is the **Network Effect**.
- **Metcalfe's Law:** Value of a network is proportional to the square of users ($n^2$).
- A new social media startup cannot compete with Facebook not because Facebook has better tech, but because Facebook has *everyone else*.
- This leads to "Winner Take All" markets.



# Part 2 Summary

- Monopolies are Price Makers.
- They produce less and charge more than competitive markets.
- This creates **Deadweight Loss** (Economic Inefficiency).
- Governments regulate them, especially in Tech where Network Effects create natural moats.
- **Next:** How do we calculate this using Python?



## Part 3: Computational Approach

# Measuring Concentration

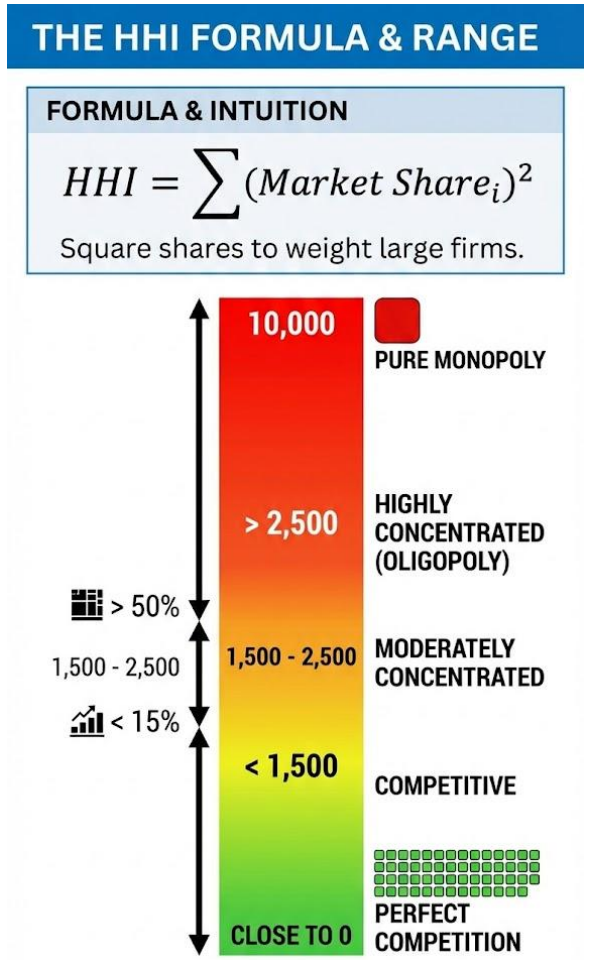
- How does a regulator decide if a market is "Too Concentrated"?
- They need a metric.
- **Concentration Ratio ($CR_n$):** Sum of market shares of top ***n*** firms.
- **Herfindahl-Hirschman Index (HHI):** A more robust metric used by the DOJ.



# The HHI Formula

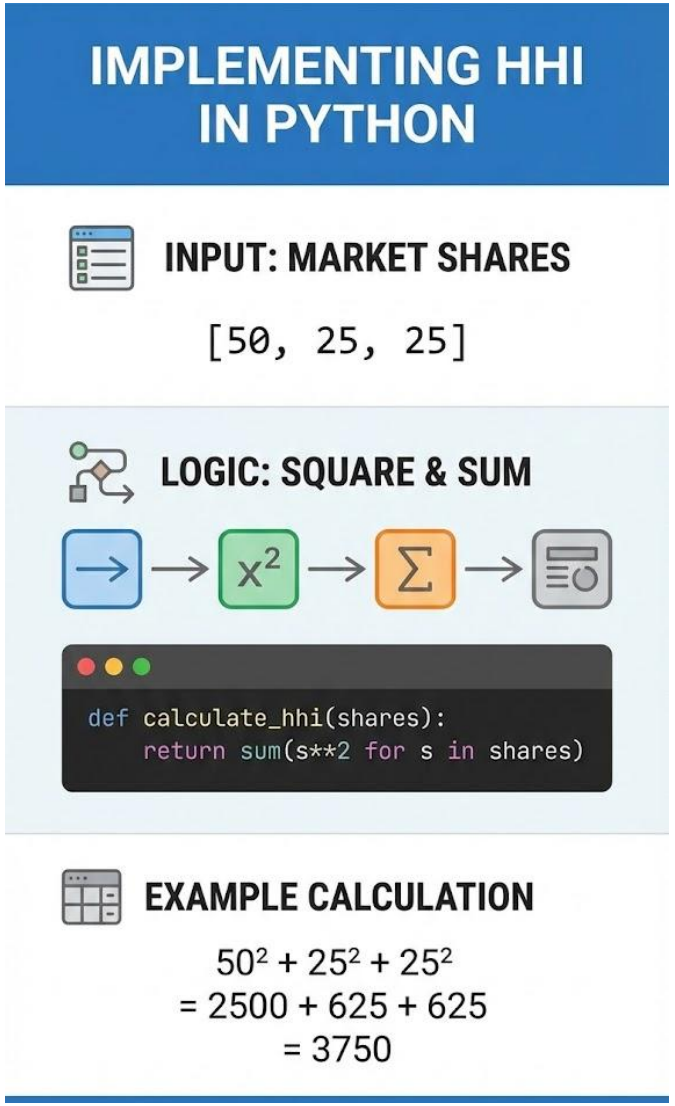
$$HHI = \sum (MarketShare_i)^2$$

- We square the market shares to give extra weight to large firms.
- **Range:**
  - **Close to 0:** Perfect Competition.
  - **10,000:** Pure Monopoly ($100^2$).
- **Thresholds:**
  - < 1500: Competitive.
  - 1500 - 2500: Moderately Concentrated.
  - > 2500: Highly Concentrated (Oligopoly).



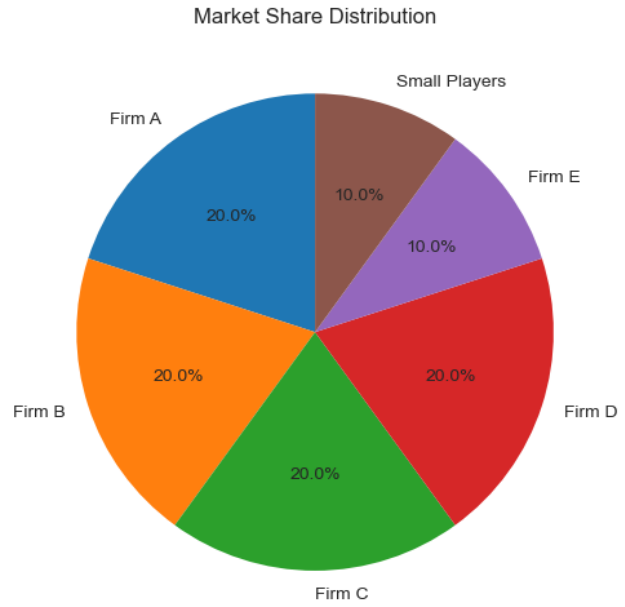
# Implementing HHI in Python

- In the tutorial, we build an hhi_calculator.
- **Input:** A list or array of market shares (e.g., [50, 25, 25]).
- **Logic:**
  - Square each number.
  - Sum them up.
- **Example:** $50^2 + 25^2 + 25^2 = 2500 + 625 + 625 = 3750$ (Highly Concentrated).



# Visualizing Market Structure

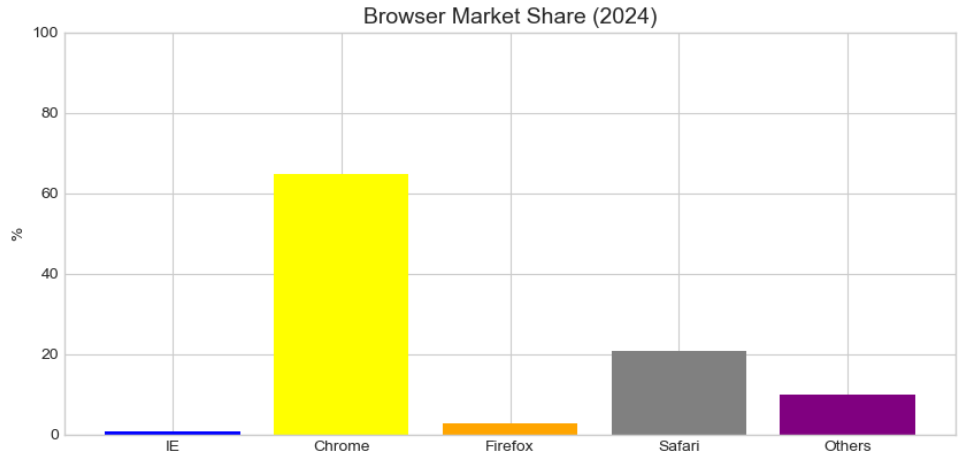
- We use matplotlib to create visualizations.
- **Pie Charts:** Good for seeing the split.
- **HHI Gauge:** Visualizes where the market sits on the regulatory spectrum.
- **Interactive Widget:** You will adjust sliders for "Firm A", "Firm B" to see how the HHI changes dynamically.

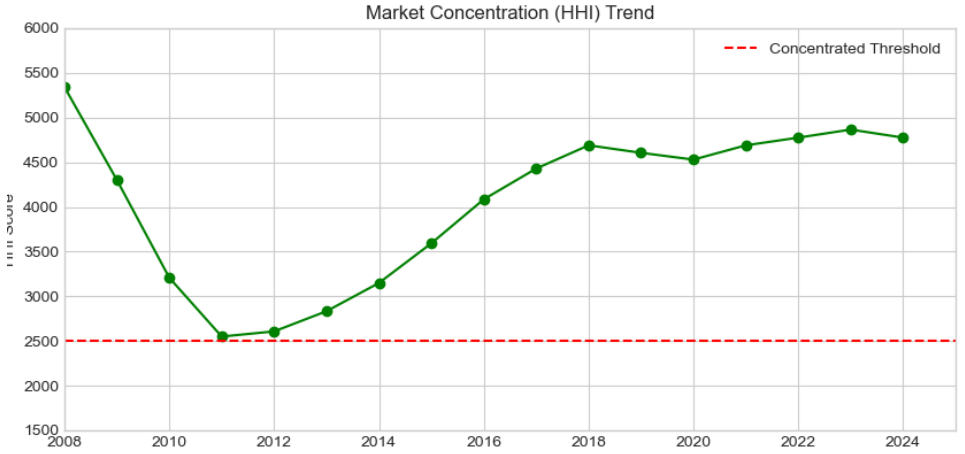





# Animating the Browser Wars

- We use historical data (IE, Chrome, Firefox) to animate market evolution.
- **Observation:**
  - 2008: High HHI (IE Dominance).
  - 2012: Low HHI (Intense Competition).
  - 2023: High HHI (Chrome Dominance).
- **Lesson:** Market structure is not static; it evolves with technology.



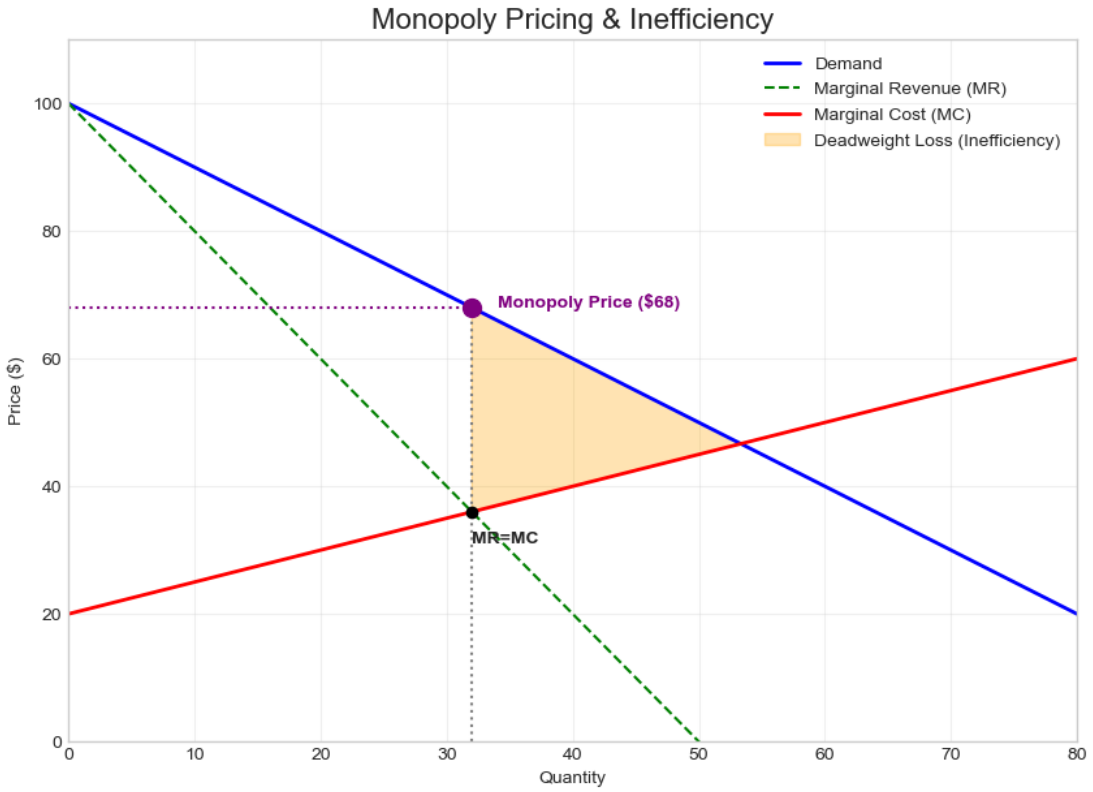
# Plotting Monopoly Profit & DWL

- We can use Python to solve for the optimal Monopoly quantity.
- **Step 1:** Define Demand Function.

  $P = a - bQ$
- **Step 2:** Define Marginal Revenue.

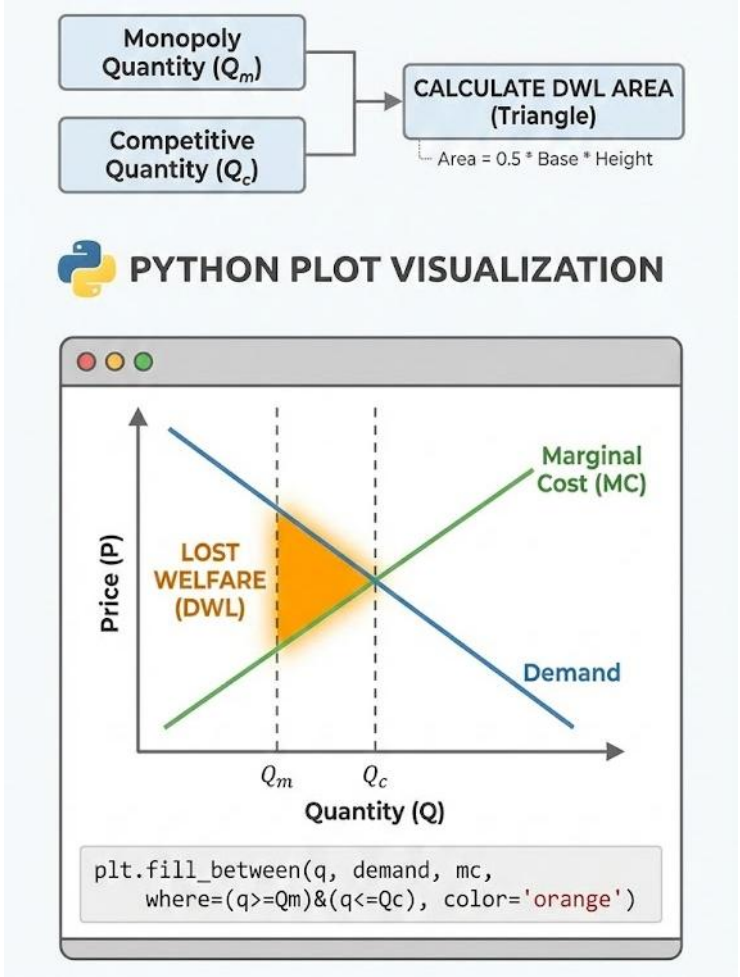
  $P = a - 2bQ$
- **Step 3:** Define Marginal Cost.

  $P = c + dQ$
- **Step 4:** Find intersection of MR and MC.



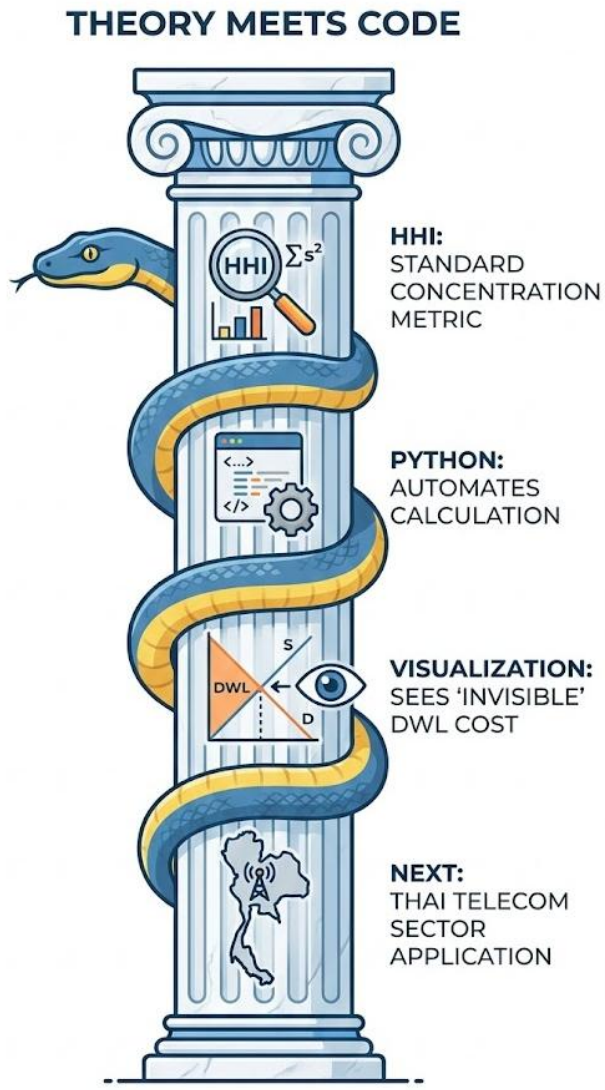
# Identifying Deadweight Loss in Code

- Once we have the Monopoly Quantity ($Q_m$) and Competitive Quantity ($Q_c$), we can calculate the area of the Deadweight Loss triangle.
- **Visualization:** We use matplotlib.pyplot.fill to shade the area between the Demand curve and MC curve, bounded by $Q_m$ and $Q_c$.



# Part 3 Summary

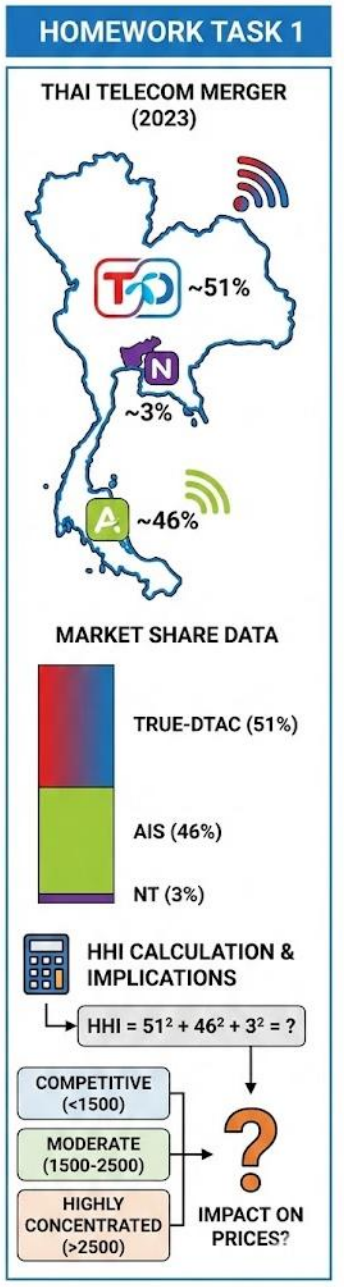
- **HHI** is the standard metric for market concentration.
- **Python** allows us to automate this calculation for any industry.
- **Visualization** helps us see the "invisible" cost of monopolies (Deadweight Loss).
- **Next:** Applying this to the Thai Telecom Sector.



Part 4: Homework Preview

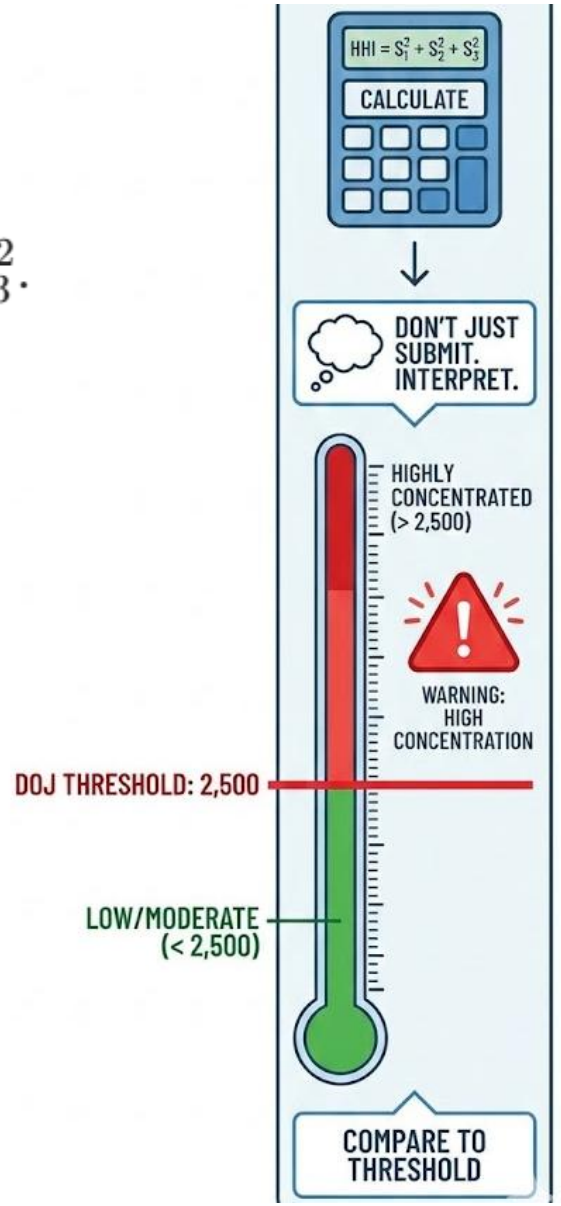
# HW Task 1: Thai Telecom Merger

- **Context:** In 2023, the Thai mobile market saw a massive merger (True & DTAC).
- **Data:**
  - Entity A (Merged): ~51%
  - Entity B (AIS): ~46%
  - Entity C (NT): ~3%
- **Task:** Calculate the HHI.
- **Question:** Is this market Competitive, Moderate, or Highly Concentrated? What does this imply for consumer prices?



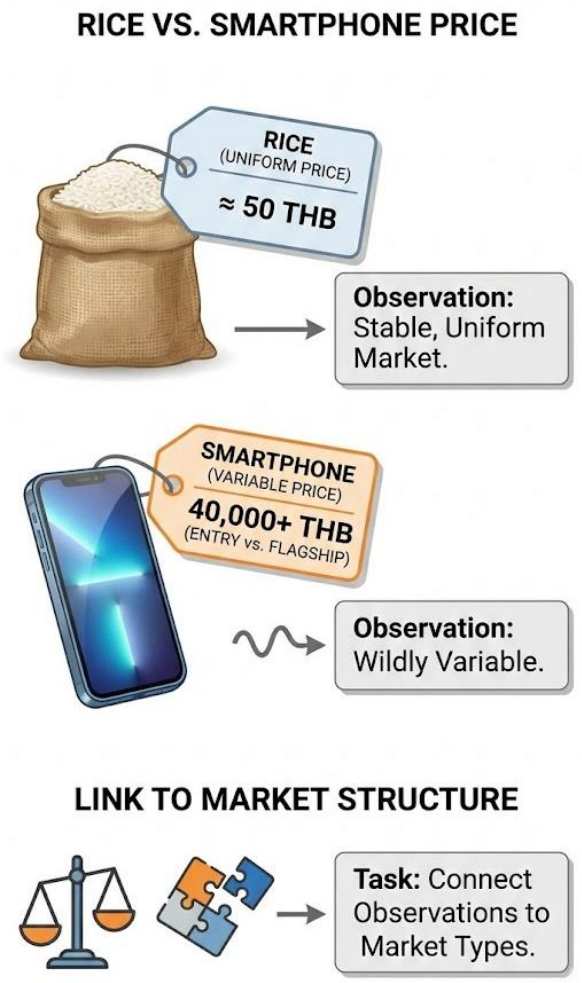
# Hint for Task 1

- Remember the formula: $HHI = S_1^2 + S_2^2 + S_3^2$.
- Don't just submit the number. **Interpret it.**
- Compare it to the DOJ threshold of 2,500.



# Homework Task 2: Analysis of Daily Decision

- **Scenario:** Compare buying Rice vs. Buying a Smartphone.
- **Observation:**
  - Rice Price: Uniform, stable.
  - Phone Price: Wildly variable (Entry level vs Flagship).
- **Task:** Link this observation to **Market Structure**.



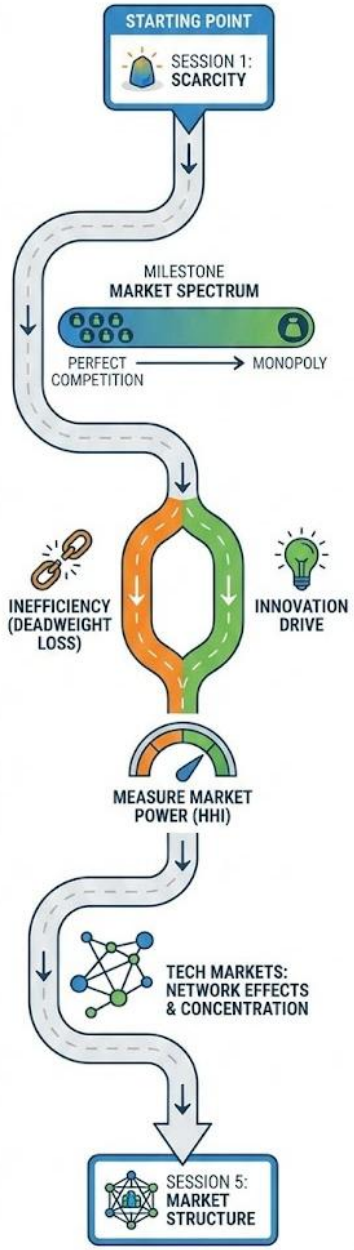
# Concept hint: Product Differentiation

- Why is the iPhone price so high?
- Is it just hardware costs? Or is it the **Ecosystem** (Monopoly power over iOS)?
- Contrast this with Rice: If one farmer doubles the price, what do you do? (Switch). If Apple doubles the price, do you switch? (Maybe not, due to switching costs).



# Final Recap

- Markets exist on a spectrum from Perfect Competition to Monopoly.
- Monopolies create inefficiency (Deadweight Loss) but may drive innovation.
- We use HHI to measure market power.
- Tech markets are prone to concentration due to Network Effects.



# Let’s check out our Notebook

